MADHUVARSU RUPESH CHANDRA BHARADWAJ

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Technical Skills

AI	Artificial Intelligence, Machine Learning, Reinforcement Learning, PPO, Neural Networks, Natural Language Processing, LLMs
Programming	Java, JavaScript, SQL
Full Stack	React.js, Node.js, Express.js, HTML5, CSS3, Tailwind CSS, REST APIs, React Query
Databases	MongoDB, Mongoose, PostgreSQL, SQL, DBMS
Deployments	GitHub, Postman, CI/CD
Core Knowledge	Data Structures, Algorithms, Software Engineering

Projects

Hybrid LLM-RL Architecture with Personality-Based NPC Behavior Repository Link

- Developed a hybrid AI system combining **PPO-based Reinforcement Learning with an LLM explanation engine**, enabling NPCs to make tactical decisions and generate human-readable reasoning.
- Designed a custom RPG combat environment with **78-dimensional observations** and four AI personalities: Aggressive, Defensive, Strategist, and Chaotic.
- Trained each personality for **250K+ timesteps** and evaluated performance using win rate, threat-targeting accuracy, and healing efficiency, achieving **75%+ win rates, 85% threat-targeting accuracy, and 80% healing efficiency**.

AI Voice-Based Assistant Repository Link

- Developed a voice-driven AI assistant combining **speech recognition, LLM-based processing, conversational context, and text-to-speech** for natural voice interaction.
- Designed an end-to-end AI workflow for converting speech into structured requests, generating contextual responses, and producing spoken output.
- Experimented with **prompt engineering and response evaluation** to improve relevance, consistency, instruction following, and conversational usability.

Car Rental – Velora Website Link Repository Link

- Built and deployed a full-stack **MERN platform** with dual dashboards, a WhatsApp-style booking chatbot, and a dynamic pricing engine supporting flat-rate and per-litre billing.
- Integrated **Razorpay** for real advance payments and developed REST APIs, JWT authentication, role-based access control (RBAC) database workflows, and responsive user interfaces.

College Discovery Platform Website Link Repository Link

- Built and deployed a full-stack **PERN application** enabling college search, comparison, community reviews, and discussions, serving **50+ users**.
- **PostgreSQL integration, REST APIs, and server-side pagination** for secure and scalable application workflows.

Education

B.Tech in Computer Science and Engineering — **VIT-AP University** — 2023 – Present
CGPA: 8.3

Intermediate (Class XII) — **Narayana Junior College, Nellore** — 2021 – 2023
Percentage: 95%